SUMMARY · MK 4 OPERATIONS & SUPPLY CHAIN MANAGEMENT · WEEK 9

Lean Six Sigma & Quality Management

This week treats quality with numbers, not slogans. Part one separates design quality from conformance quality (TQM), then the four cost-of-quality categories (prevention, appraisal, internal failure, external failure) and the "$1 of prevention saves $10" logic. Part two explains Six Sigma: DPMO, the 3.4 defects per million target, the sigma scale, the DMAIC cycle, analysis tools (Pareto, fishbone, control charts, FMEA, poka-yoke), the seven wastes, and belt and champion roles. Part three compares Lean (flow) and Six Sigma (variation), why they combine, and Baldrige vs ISO 9000. It closes with the SMRT, CSSC and Pelindo cases: who pays for quality, why the CPPI rank fell, and why improvement programmes often die in the Control phase.

Material

1 TQM: two goals — design quality and conformance quality

Total Quality Management (TQM) = managing the whole organisation to excel on everything the customer values. Chapter 12 separates two goals:

Goal 2 · Conformance quality"Do we make it right, every time?" A *daily, tactical* job owned by operations: specs met on every unit, every shift.

Fitness for use needs both. A great design with poor conformance still fails (a luxury car that keeps breaking down). Perfect conformance to the wrong design also fails (a consistent product nobody wants).

Customer focus
Start from what the customer values, not what the factory finds easy to measure.

Continuous improvement
Based on process data, not slogans like "quality is our priority".

This week's goal: to argue for a quality budget with numbers — naming the cost category, the DMAIC phase and who pays.

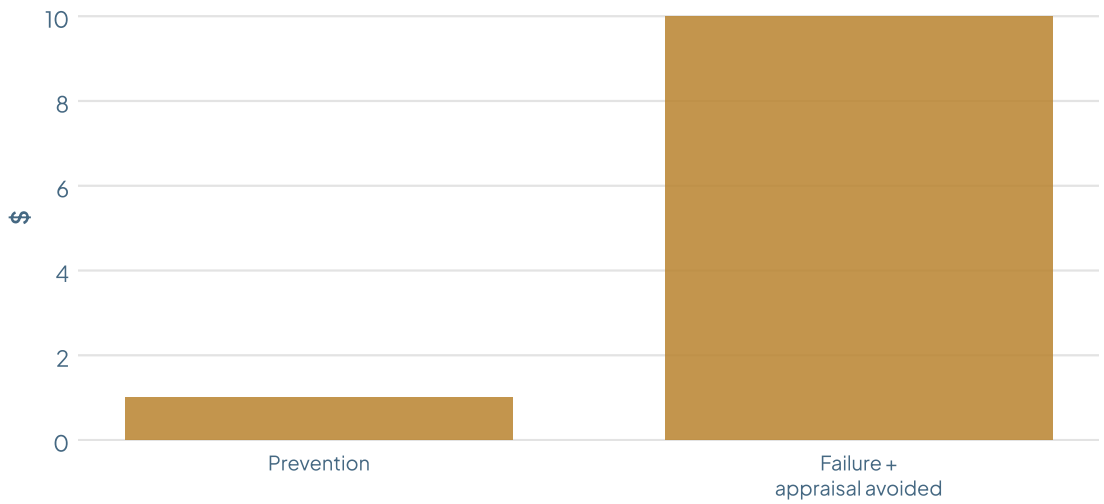
2 Cost of quality: four categories and the 1 : 10 leverage

Juran (1951) gave quality managers numbers so they could talk to the finance board. A typical cost of quality is **15–20% of sales**; in a well-run programme it can be **< 2.5%**.

Category	Role	Examples
Prevention	Find causes, fix, train, redesign. The one that pays back.	Root-cause analysis, operator training, process redesign, poka-yoke
Appraisal	Inspect and test. Finds defects, does not stop them.	Incoming inspection, final testing, quality audits

Internal failure	Defects caught before the customer sees them.	Scrap, rework, repair, downtime
External failure	Defects that reach the customer.	Warranty, complaints, lost customers and goodwill

The 1 : 10 leverage (slide illustration)



$1 on cause analysis, corrective action, redesign and training avoids about $10 of scrap, rework, warranty, lost customers and inspection.

$1 of prevention saves $10 in failure and appraisal costs. This assumes failures do occur, prevention is cheaper, and performance can be measured.

! **Why do firms still ignore it?**
(1) *Prevention is invisible* — its success is a breakdown that never happens, and nobody reports it. (2) *Appraisal is easy to count* — inspections done and units tested are easy to show in a budget meeting. (3) So the 1 : 10 ratio never bites: if you can't measure the cost of quality, you can't prove prevention is cheaper. This was CSSC's problem: its cost of quality was argued at 2% vs 30%, so nobody could prove prevention pays.

3 Six Sigma: DPMO, the 3.4 target and the sigma scale

Six Sigma is a statistical term turned into a management discipline. Jack Welch (GE): *"The big myth is that Six Sigma is about quality control and statistics. It is that — but it's much more."*

3.4
defects per million opportunities (DPMO)

6σ
how far the process is from the defect limit

99.99966%
of output within spec at Six Sigma

15–20%
of sales lost to cost of quality before improving

```
DPMO = Number of defects ÷ (Units × Defect opportunities per unit) × 1,000,000
```

≡ **Worked example (Chapter 12, Problem 10)**
2,000 units inspected × 5 criteria = 10,000 opportunities. 33 defects found. DPMO = 33 ÷ 10,000 × 1,000,000 = **3,300** → about **4σ** (more precisely about 4.2σ).

Level	Yield	DPMO (approx.)	Picture
2σ	69.1%	308,500	Manual work, no controls
3σ	93.3%	66,800	Competent, routine inspection
4σ	99.38%	6,200	Problem 10 example (3,300)
5σ	99.977%	230	Needs designed-in prevention
6σ	99.99966%	3.4	The target

Each sigma step is a giant leap — from 4σ to 5σ, defects fall about 27-fold.

! **Change the denominator, change the score**
The same 33 defects: counted per criterion → 3,300 DPMO; counted per unit (2,000 opportunities) → **16,500** DPMO. The process did not change at all. Always ask how "opportunities" are defined before comparing sigma scores.

4 DMAIC: the five-phase improvement cycle

DMAIC is for an **existing process** whose problem is known but whose cause is not.

Phase	Question	Content	Tools
Define	What is the problem?	Write the charter, listen to the customer	Charter, voice of customer, SIPOC
Measure	How big is it?	Map the process, set the baseline	As-is map, process walk, 8 wastes
Analyze	Why does it happen?	Find the root cause — causes, not solutions. *The most skipped phase.*	Fishbone, 5 Whys, Pareto
Improve	Does the fix work?	Pilot the change, verify with data before rollout	5S, pilot, to-be map
Control	Will it hold?	Monitor, plan the response, hand off to the owner	Visual management, monitoring, handoff

DMAIC is a cycle, not a checklist: Control feeds the next Define. Skip Analyze and you solve the wrong problem.

Where SMRT stopped: MEASURE

Counted 21 dislodged claws, 13 damaged trains, 34 damaged covers — *what* broke, but not *why*.

Why Analyze gets skipped

Looking for causes looks like delay next to "doing something", so teams jump ahead.

Where CSSC stopped: CONTROL

Kaizen, JIT, cells, Baldrige: each improved, then was dropped when attention moved on.

5 Analysis tools, the seven wastes, and poka-yoke

Pick the tool by the question you are asking:

Tool	Answers

Process map / VSM	What are the steps? Which add value?
Pareto chart	Which few categories carry most problems? (often ~20% of causes → ~80% of problems)
Fishbone (Ishikawa)	What could cause it? Machine, method, material, people.
Checksheet & run chart	How often, over time? The cheapest evidence.
Control chart / SPC	Did the process *change*, or just vary normally?
FMEA	Severity × occurrence × detection: what to fix first?
Design of experiments	Test many variables at once for cause and effect.
Poka-yoke	Make the error impossible, so no defect is made.

```
FMEA: Risk Priority Number (RPN) = Severity × Occurrence × Detection (each scored 1–10)
```

The seven wastes (TIMWOOD) — *muda*: anything the customer would not pay for:

T · Transport
Moving things the customer never asked to move.

I · Inventory
Stock held against uncertainty; it hides variation.

M · Motion
Reaching, searching, walking that adds nothing.

W · Waiting
Idle time between steps.

O · Over-processing
Doing more than the spec needs.

O · Overproduction
Making more, or sooner, than demand; creates the rest.

D · Defects
Rework, scrap, repair.

Where they meet
Defects are Lean's 7th waste and Six Sigma's first metric.

Shingo: inspection arrives too late. Errors are inevitable; defects are not, if feedback is immediate.

Poka-yoke model Error attempted → device blocks it → instant signal → fixed at source → no defect. Window = zero.

Applied to SMRT

A claw checked weekly = the top model: up to 7 days of risk. A claw that cannot come loose = the bottom model. The cable tie its maker recommended was a poka-yoke — and it was deferred for a study.

6 Roles, belts, and the champion problem

Six Sigma is a *people* practice before it is a statistical one.

Role	Job

Executive leaders	Truly committed. Without them, belts have authority over nothing.
Champions	Own the process, remove barriers, tie work to results.
Master black belts	Deep method; support many teams.
Black belts	Lead a team; must prove the fix is permanent.
Green belts	Join teams or run small projects in their own job.

The $17,000 sorting table (CSSC)

A good champion asked for the data, agreed, and got sign-off in **one week** — not months. Lesson: champions remove barriers and show management is aligned.

7 Lean vs Six Sigma, and why they combine

	LEAN · speed & flow	SIX SIGMA · quality & consistency
Focus	Remove waste (muda)	Reduce variation
Goal	Faster flow, shorter cycle time	Near-perfect: < 3.4 DPMO
Method	PDCA & kaizen: small, frequent steps	DMAIC: structured, data-driven
Tools	5S, VSM, kanban, JIT, SMED, standard work	Control charts, fishbone, FMEA, ANOVA, DOE
Symptoms	Queues, waiting, long lead time, short touch time	Output drifts; results differ by shift or site

Quick test: listen to the customer's complaint

"**It takes too long**" → a flow problem → **Lean**. "**It's not the same every time**" → a variation problem → **Six Sigma**.

Why they combine (Lean Six Sigma): less inventory needs more reliable processes. Lean removes the buffer; Six Sigma makes the process reliable enough to work without it.

Six Sigma alone Consistent output — possibly consistently making things nobody ordered, or with long lead times.

8 Baldrige vs ISO 9000

	Malcolm Baldrige Award	ISO 9000
Question	How good are your **results**?	Do you **do what you say you do**?
Run by	NIST, US Department of Commerce	International standard, adopted by the EU
Form	Up to 18 awards a year; 7 categories from leadership to results; scored out of 1,000 (~650+ earns a site visit)	Third-party registrars audit regularly; certifies the *system*, not the product

! Important

You can be ISO certified and still make a mediocre product — consistently. Certification moves a firm up (documented system), but not to the right (better results).

↕ System: documented ↑

ISO 9000 certified
Documented and audited, but the spec may be mediocre.

Baldrige calibre
Documented and strong results in all 7 categories.

Where CSSC sits
Manual, SPC charts, 0.25% rejects — but $1.48M to get there.

Good outcomes, fragile
Results hold while key people stay.

↔ Results: weak → strong

For CSSC: its standard is met (customers reject only 0.25% of parts); the problem is the **cost**, $1.48M. Four fixes were dropped at the Control phase. A certificate documents today's process — it won't improve it, and uses the same scarce capacity. Also in Chapter 12: ISO 14000 (environment), ISO 26000 (social responsibility), QS-9000 (auto suppliers), and benchmarking (learning from the best).

9 The Pelindo case: the score and the limits of control

Rangga Pradipta, SVP Operational Excellence (a composite character): "For four years we improved what we could measure. Were we measuring the right thing?"

+90%
Faster cranes at Belawan: 20 → 38 boxes per ship per hour (BSH)

-42%
Less ship time at Belawan: 55 → 32 hours in port

23 → 121
Tanjung Priok world rank (CPPI 2023 → 2025 edition)

95%
Pelindo's container market share — customers can't go elsewhere

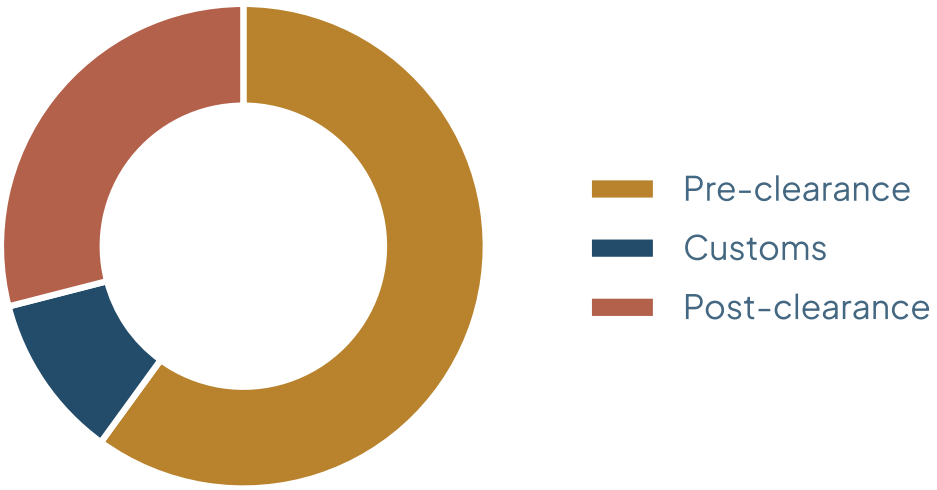
Standardisation worked: the same procedures at every port, one IT system (PTOS), a central control room, equipment moved where needed, operator training. This is *prevention* money that worked — Pelindo reports IDR 1.3 trillion of post-merger efficiency.

Metric	Meaning	Owner	Status

BSH	Boxes per ship per hour (crane speed)	Pelindo	Rose at every port
Port stay	Hours a ship spends in port	Mostly Pelindo	Down 35–67%
Dwelling time	Days cargo waits until it leaves the gate	Many agencies	2.9 days; target 2.85 (2026)
CPPI rank	World Bank/S&P Global index of ship time in port, from 10 global lines	Nobody in Indonesia	Rank 121 (2025)

Pelindo fully controls the number it is best at, and has no control over the number the world reads.

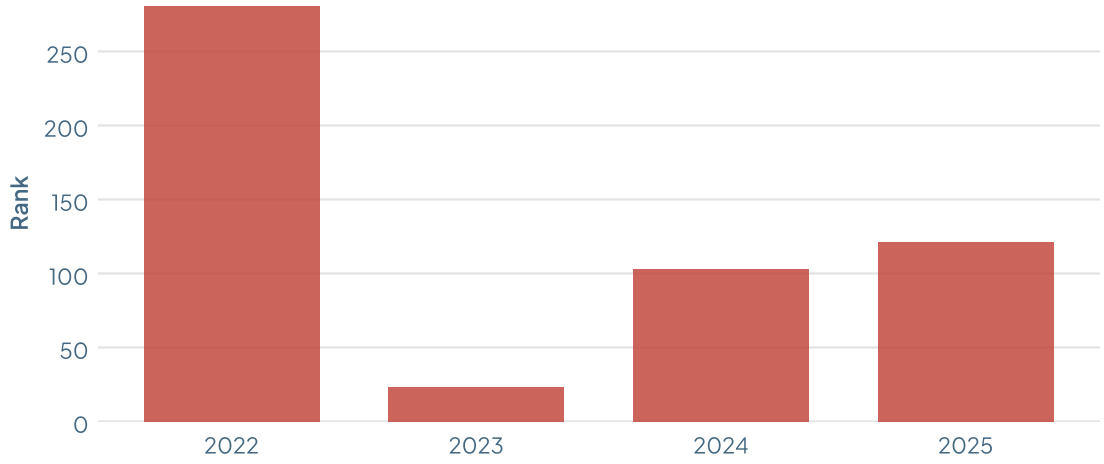
Tanjung Priok import dwelling time by stage (2015 study)



Pre-clearance 60% (importer papers, permits, DO — no Pelindo control); customs 11% (green/yellow/red channel — none); post-clearance 29% (gate, yard, trucks — partial). Faster cranes cannot fix late papers.

Check the ruler before you read the number. CPPI counted only 834 port calls (2022), while Tanjung Priok serves 8,000+ calls a year (about 1 in 10), from 10 global lines that rarely call there. Logistics cost fell from 23.5% to 14.29% of GDP mainly because the *formula* changed, not because costs really fell. When a number jumps, first ask: "Is this still the same ruler?"

Tanjung Priok CPPI rank (lower is better)



281 → 23 → 103 → 121, though crane speed didn't drop and volume grew. Part of the swing comes from method changes and a small sample (834 of 8,000+ calls).

10 Answers to the Pelindo discussion questions

The answers below are our analysis of the slide questions, not an official answer key.

Q1 · Who pays for quality? Standardisation, PTOS and training are **prevention**. Red-channel checks, behandle and scanning are **appraisal** — they find problems, they don't prevent them. The biggest failure cost is **external failure**: demurrage and cargo waiting time, paid by *cargo owners*, not Pelindo. Because that bill never hits Pelindo's books, "Rp1 of prevention saves Rp10" doesn't bite for Pelindo. With a 95% market share, unhappy customers can't leave — the only warning signals left are complaints and international rankings, and they arrive late.

Q2 · Why did the rank fall 98 places? Use Analyze-phase logic. *Symptom*: rank 23 → 103 → 121. *Not a root cause*: Pelindo's operations — crane speed didn't drop, ship time didn't rise, volumes grew. The 2023 jump to 23 had three causes (better operations, post-Covid recovery, a new counting method), and Pelindo did only one. The denominator is small (834 of 8,000+ calls) and from lines that rarely call. If the ruler changes but the process doesn't, a control chart still shows a shift — a shift in *measurement*, not the process.

Q3 · Too long, or not the same? Both exist. "Cargo takes too long" = a flow problem → **Lean** (and mostly outside the terminal fence: 60% pre-clearance, 11% customs). "Service is not the same at every port" (BSH Sorong 25, Nilam 35, Belawan 38) = a variation problem → **Six Sigma**. A higher average BSH is not a narrower spread — ask for the distribution by port and shift. To avoid dying in Control like CSSC's four programmes, live monitoring (the control room) must be joined by a **response plan** and a **named owner**.

If the root cause sits in another organisation, DMAIC still works, but the Define phase must change the project boundary: bring the other agencies (customs, ministries, shipping lines) in as co-owners, and choose a metric that reflects the customer's experience (end-to-end dwelling time), not just internal metrics.

How to argue well (from the slides)
(1) Name the cost category, (2) name the DMAIC phase, (3) name who pays, (4) say what would change your mind.

11 Six takeaways from the three cases

1. Prevention works
Pelindo's standardisation raised BSH at every port it touched.

2. Improvement stops where the bill disappears
CSSC paid its own $1.48M. Pelindo never sees its external-failure bill.

3. Inspection is not prevention
Patrols, end-of-line checks and red-channel checks find defects — they don't make fewer.

4. Check the denominator
834 vs 8,000 calls; 23.5% vs 14.29%; 2% vs 30%. Same ruler? Do you own it?

5. Lean for flow, Six Sigma for variation
"Too long" is flow. "Not the same" is variation. Pelindo has both.

6. Control is where programmes die
Monitoring is not enough. You also need a response plan and a named owner.

Key takeaways

1. TQM has two goals: design quality (strategic) and conformance quality (operational, every unit every shift).
2. Four quality costs: prevention, appraisal, internal failure, external failure. $1 of prevention saves about $10.
3. Typical cost of quality is 15–20% of sales; a good programme < 2.5%.
4. DPMO = defects ÷ (units × opportunities) × 1,000,000. Six Sigma = 3.4 DPMO (99.99966%). The definition of opportunity changes the score.
5. DMAIC: Define, Measure, Analyze, Improve, Control. Analyze is skipped most; Control is where programmes die.
6. Pick tools by question: Pareto (vital few), fishbone (causes), control chart (changed or just varying), FMEA (RPN = S × O × D), poka-yoke (prevent at source).
7. TIMWOOD: transport, inventory, motion, waiting, over-processing, overproduction, defects.
8. Lean = flow ("too long"); Six Sigma = variation ("not the same"). Combined because low stock needs reliable processes.
9. Baldrige judges results; ISO 9000 checks you do what you say you do.
10. Pelindo: BSH and port stay improved, but dwelling time and CPPI are controlled by others; check the ruler before reading the number.

Case studies

Worked examples

Problem 1 · Easy **DPMO and the definition of opportunity**

A plant inspects 5,000 parts; each is judged on 4 criteria. 60 defects are found. Compute DPMO with opportunity = criterion, then with opportunity = unit.

1. Opportunities (per criterion) = 5,000 × 4 = 20,000.

```
DPMO = 60 ÷ 20,000 × 1,000,000 = 3,000
```

2. Opportunities (per unit) = 5,000 → DPMO = 60 ÷ 5,000 × 1,000,000 = **12,000**.

3. 3,000 (about 4.2σ) looks far better than 12,000 (about 3.8σ), yet the process is the same. Agree on the opportunity definition before comparing.

Answer: 3,000 DPMO (per criterion) vs 12,000 DPMO (per unit).

Problem 2 Medium **Classifying the cost of quality**

One year's data (IDR million), sales 10,000: SPC training 120; final inspection 300; rework 450; scrap 250; warranty claims 600; mould redesign 80; incoming material tests 150. Classify, compute cost of quality as % of sales, and suggest where the budget should go.

1. Prevention = training 120 + redesign 80 = **200**.
2. Appraisal = final inspection 300 + incoming tests 150 = **450**.
3. Internal failure = rework 450 + scrap 250 = **700**. External failure = warranty **600**.
4. Total = 200 + 450 + 700 + 600 = 1,950 → **19.5%** of sales (in the typical 15–20% range).
5. Only about 10% goes to prevention while 67% goes to failure. Shift budget to prevention (root-cause analysis, poka-yoke) and track the fall in failures.

Answer: P 200, A 450, IF 700, EF 600; total IDR 1,950m = 19.5% of sales.

Problem 3 Medium **FMEA priority with RPN**

Three failure modes (1–10 scales): (A) claw dislodged: severity 10, occurrence 3, detection 7. (B) cover cracked: 6, 6, 3. (C) loose bolt: 7, 4, 5. Rank the priorities.

1. `RPN = Severity × Occurrence × Detection`
2. A = 10 × 3 × 7 = **210**; B = 6 × 6 × 3 = **108**; C = 7 × 4 × 5 = **140**.
3. Order: A → C → B. Note: a severity of 10 (safety) is often tackled first whatever the RPN. Poka-yoke lowers occurrence; better inspection only lowers the detection score.

Answer: A (210), C (140), B (108).

Practice questions

Try answering first. The answer key is below each question.

1. "Do we make it to spec, every unit, every shift?" is a question of…

A. Design quality
B. Conformance quality
C. Perceived quality
D. Aesthetics

Answer: B. Conformance quality is operations' daily job.

2. Operator training and process redesign are which cost?

A. Appraisal
B. Prevention
C. Internal failure
D. External failure

Answer: B. Prevention finds and removes causes before defects happen.

3. Warranty claims and lost customers are which cost?

A. Prevention
B. Appraisal
C. Internal failure
D. External failure

Answer: D. The defect reached the customer — external failure, usually the most expensive.

---

4. Why do firms often under-invest in prevention?

A. Prevention is always more expensive
B. Prevention's success is invisible and cost of quality is hard to measure
C. ISO forbids it
D. Appraisal works better

Answer: B. Breakdowns that never happen aren't reported; appraisal is easy to count in budget meetings.

---

5. 2,000 units × 5 criteria, 33 defects found. DPMO is…

A. 330
B. 3,300
C. 16,500
D. 33,000

Answer: B. 33 ÷ 10,000 × 1,000,000 = 3,300 (about 4σ).

---

6. Six Sigma means about … defects per million opportunities.

A. 0
B. 3.4
C. 233
D. 6,210

Answer: B. 3.4 DPMO = 99.99966% within spec.

---

7. The same 33 defects counted per unit (2,000 opportunities) give a DPMO of…

A. 3,300
B. 6,600
C. 16,500
D. 33,000

Answer: C. 33 ÷ 2,000 × 1,000,000 = 16,500. Same process — different denominator.

---

8. The DMAIC phase most often skipped is…

A. Define
B. Measure
C. Analyze
D. Control

Answer: C. Looking for causes looks like delay, so teams jump to solutions.

---

9. SMRT counted dislodged claws and damaged trains but never asked why. It stopped at…

A. Define
B. Measure
C. Improve
D. Control

Answer: B. Measure answers "how big", not "why".

---

10. The tool to tell whether a process really changed or just varies normally is…

A. Pareto chart
B. Control chart / SPC
C. 5S
D. Kanban

Answer: B. Control charts separate common variation from real shifts.

---

11. Severity 8, occurrence 5, detection 6. The FMEA RPN is…

A. 19
B. 48
C. 240
D. 480

Answer: C. 8 × 5 × 6 = 240. Highest RPN is fixed first.

---

12. The waste the slides say "creates the rest" is…

A. Motion
B. Overproduction
C. Transport
D. Over-processing

Answer: B. Making more or sooner than demand triggers stock, transport, waiting and hidden defects.

---

13. A cable tie that makes SMRT's claw impossible to dislodge is an example of…

A. Appraisal
B. Poka-yoke
C. Benchmarking
D. ISO 9000

Answer: B. Poka-yoke makes the error impossible, so the risk window = zero.

---

14. Customers complain "it's not the same every time". The main approach is…

A. Lean
B. Six Sigma
C. ISO 14000
D. Kanban

Answer: B. Not the same = variation → Six Sigma. Too long = flow → Lean.

---

15. Why are Lean and Six Sigma combined?

A. To earn two certificates
B. Lean removes the buffer; Six Sigma makes the process reliable without it
C. Six Sigma replaces kaizen
D. They are identical

Answer: B. Low inventory needs reliable processes.

---

16. ISO 9000 essentially checks…

A. How good your results are
B. Whether you do what you say you do
C. Product design quality
D. Cost of quality

Answer: B. ISO certifies the system; Baldrige judges results.

---

17. In the Pelindo case, the metric Pelindo controls most is…

A. CPPI rank
B. National dwelling time
C. BSH (boxes per ship per hour)
D. Logistics cost % of GDP

Answer: C. BSH is crane speed at the berth — fully Pelindo's.

---

18. Demurrage from cargo waiting too long, paid by cargo owners, is which cost?

A. Prevention
B. Appraisal
C. Internal failure
D. External failure

Answer: D. A failure felt by the customer; it never hits Pelindo's books, so the 1 : 10 logic doesn't bite.

Sources

- P3MD Week 9 MK4 — Lean Six Sigma & Quality Management (incl. SMRT, CSSC, Pelindo cases)
- Course OSCM textbook, Chapter 12 — Quality Management / Six Sigma